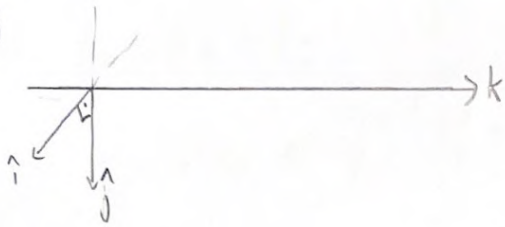


3)



Infinite length string under tension (constant) T and homogeneous mass density μ .

We have found in class transverse displacements (perpendicular to $\hat{k}$) on a unique plane obeys:

$$\mu \frac{\partial^2 \vec{f}}{\partial t^2} = T \frac{\partial^2 \vec{f}}{\partial z^2} \quad (\text{call this the wave equation})$$

→ Due to linearity in this equation we can capture the displacements in any direction in the $\hat{i}$ - $\hat{j}$ plane.

$$\vec{f} = f_x \hat{i} + f_y \hat{j} \quad \mu \frac{\partial^2 f_x}{\partial t^2} \hat{i} + \mu \frac{\partial^2 f_y}{\partial t^2} \hat{j} = T \frac{\partial^2 f_x}{\partial z^2} \hat{i} + \frac{\partial^2 f_y}{\partial z^2} \hat{j}$$

Although not the only one sinusial waves are solutions (seperately) to the wave equation. In fact any geometrical form that is travelling in time dimension with $v = \sqrt{\frac{T}{\mu}}$ is a solution.

Let us propose a solution as:

$$\vec{f} = f_x^0 \sin(kz - \omega t + \phi_x) \hat{i} + f_y^0 \sin(kz - \omega t + \phi_y) \hat{j}$$

$$\frac{\partial^2 \vec{f}}{\partial t^2} = -f_x^0 \omega^2 \sin(kz - \omega t + \phi_x) \hat{i} - f_y^0 \omega^2 \sin(kz - \omega t + \phi_y) \hat{j} = -\omega^2 \vec{f}$$

$$\frac{\partial^2 \vec{f}}{\partial z^2} = -f_x^0 k^2 \sin(kz - \omega t + \phi_x) \hat{i} - f_y^0 k^2 \sin(kz - \omega t + \phi_y) \hat{j} = -k^2 \vec{f}$$

$$\hookrightarrow \text{Clearly} \quad k^2 \frac{\partial^2 \vec{f}}{\partial t^2} = \omega^2 \frac{\partial^2 \vec{f}}{\partial z^2} \Rightarrow \frac{\partial^2 \vec{f}}{\partial t^2} = v^2 \frac{\partial^2 \vec{f}}{\partial z^2} \Rightarrow \mu \frac{\partial^2 \vec{f}}{\partial t^2} = T \frac{\partial^2 \vec{f}}{\partial z^2}$$

If $\phi_x = \phi_y = \phi$, $\vec{f}$ turns into

$$\vec{f} = \sin(kz - \omega t + \phi) [f_x^0 \hat{i} + f_y^0 \hat{j}] \Rightarrow \text{this clearly defines a constant plane}$$

* Each particle on the string will oscillate in the $f_x^0 \hat{i} + f_y^0 \hat{j}$ direction with frequency ω . That is to say the whole wave propagates on a particular.

If $\phi_x \neq \phi_y$, individual waves (displacements to be politically correct) in $\hat{i}$ & $\hat{j}$ direction are not in phase. Therefore particles on the string will no more oscillate in one particular plane.

→ First assume $\phi_y = \phi_x + \frac{\pi}{2}$, then $\vec{f}$ transforms into

$$\vec{f} = f_x^0 \sin(kz - \omega t + \phi_x) \hat{i} + f_z^0 \cos(kz - \omega t + \phi_x) \hat{j}$$

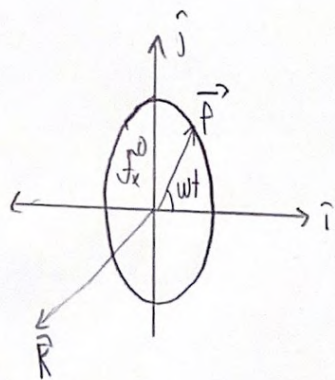
If we look at the string at a specific time $t = t_0$, we will see an ellipse with axes f_x^0 & f_z^0

When we focus on a specific segment of the segment $z = z_0$

$$\vec{f} = f_x^0 \sin(kz_0 - \omega t + \phi_x) \hat{i} + f_z^0 \cos(kz_0 - \omega t + \phi_x) \hat{j}$$

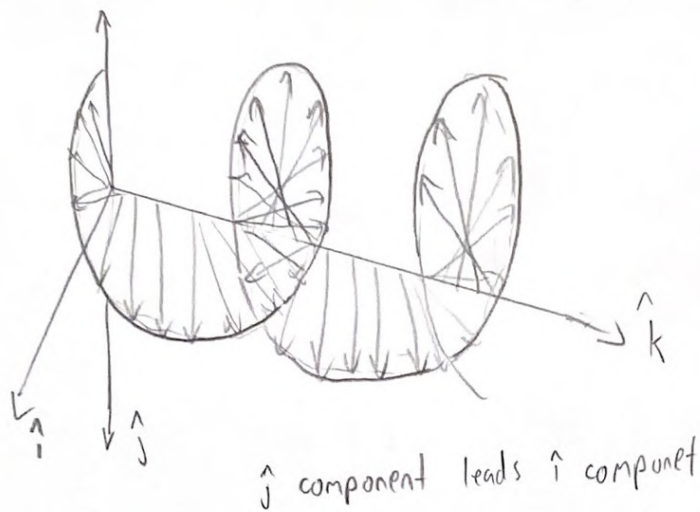
↳ displacement vector ^{particles} will clearly trace on ellipse with axes f_x^0 and f_z^0 in general

If $f_x^0 = f_z^0$, it will be a circle.



* Also notice that projection of $\vec{f}$ onto $\hat{i}$ & $\hat{j}$ axes will be sinusoidal waves we've found earlier.

Our observations begs for a wave of the form:



* This is a right circularly polarized wave since y component is $\pi/2$ ahead of x

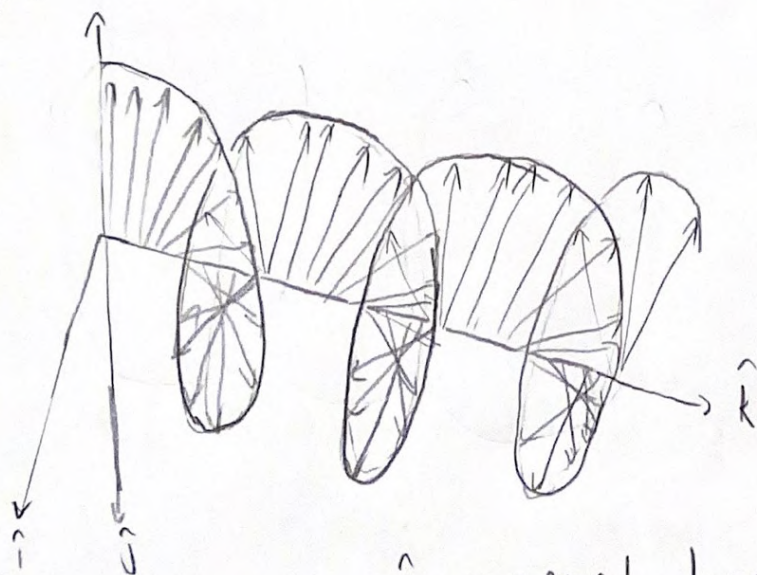
→ If you twist your fingers along the spiral, the direction your thumb points is the propagation direction.

↳ RIGHT HANDEDLY

Similarly if $\phi_y = \phi_x - \frac{\pi}{2}$ one has:

$$\vec{E} = E_x^0 \sin(kz - \omega t + \phi_x) \hat{i} - E_y^0 \cos(kz - \omega t + \phi_y) \hat{j}$$

All of the arguments for the before case is still pertinent, but this time y component is $\pi/2$ behind the x component. Therefore it is a left-handed circularly polarized wave. You have to twist your left hand in order to find the direction of propagation.



j component lags i component

4)

$$y(x,t) = A^+ \sin(kx - \omega t) + A^- \sin(-kx - \omega t)$$

Homogeneous string under tension T and mass density μ .

Solution can be provided as a superposition of a right-moving $(+\hat{x})$ and left-moving $(-\hat{x})$ sinusoidal waves.

$$\mathcal{E} = \frac{1}{2} \mu \dot{y}^2 + \frac{1}{2} T y'^2 \quad \& \quad \mathcal{J} = -T \dot{y} y'$$

$$\dot{y} = -A^+ \omega \cos(kx - \omega t) - A^- \omega \cos(-kx - \omega t)$$

$$y' = +A^+ k \cos(kx - \omega t) - A^- k \cos(-kx - \omega t)$$

$$\begin{aligned} \mathcal{E} &= \frac{1}{2} \mu \omega^2 \left[A^{+2} \cos^2(kx - \omega t) + 2A^+ A^- \cos(kx - \omega t) \cos(-kx - \omega t) + A^{-2} \cos^2(-kx - \omega t) \right] \\ &+ \frac{1}{2} T k^2 \left[A^{+2} \cos^2(kx - \omega t) - 2A^+ A^- \cos(kx - \omega t) \cos(-kx - \omega t) + A^{-2} \cos^2(-kx - \omega t) \right] \end{aligned}$$

We know that $v^2 = \frac{T}{\mu}$ and $\omega^2 = v^2 k^2$. The $\mathcal{E}$ becomes

$$\begin{aligned} \mathcal{E} &= \frac{1}{2} \mu \omega^2 \left[2A^{+2} \cos^2(kx - \omega t) + 2A^{-2} \cos^2(-kx - \omega t) \right] \\ &= \mu \omega^2 \left[A^{+2} \cos^2(kx - \omega t) + A^{-2} \cos^2(-kx - \omega t) \right] \end{aligned}$$

$$\begin{aligned} \mathcal{J} &= -T \left[-\omega k A^{+2} \cos^2(kx - \omega t) + \omega k A^{-2} \cos^2(-kx - \omega t) \right. \\ &\quad \left. + \omega k A^+ A^- \cos(kx - \omega t) \cos(-kx - \omega t) - \omega k A^+ A^- \cos(kx - \omega t) \cos(-kx - \omega t) \right] \\ &= +T \omega k \left[A^{+2} \cos^2(kx - \omega t) - A^{-2} \cos^2(-kx - \omega t) \right] \end{aligned}$$

5) Consider a system consisting of N generalized coordinates q_i & their respective velocities $\dot{q}_i$ with a Lagrangian

$$L = \sum_i \sum_j \frac{1}{2} M_{ij} \dot{q}_i \dot{q}_j - \sum_i \sum_j \frac{1}{2} V_{ij} q_i q_j$$

where M & V (non-singular) are symmetric matrices with $M^T = M, V^T = V$ (Also M & V are positive eigenvalue matrices)

We are given a new set of generalized co-ordinates $\tilde{q}_i$ with:

$$L = \frac{1}{2} \sum_{\alpha} M_{\alpha} \dot{\tilde{q}}_{\alpha} \dot{\tilde{q}}_{\alpha} - \frac{1}{2} \sum_{\alpha} \sum_{\beta} \tilde{V}_{\alpha\beta} \tilde{q}_{\alpha} \tilde{q}_{\beta}$$

Now assume in it's most general form a transform into $\tilde{q}$ with an orthonormal rotation (in generalized co-ordinate space) via an orthonormal matrix and a translation. That is:

$$\left. \begin{aligned} \tilde{q}_{\alpha} &= O_{\alpha i} q_i + a_{\alpha} \\ \tilde{q}_{\beta} &= O_{\beta j} q_j + a_{\beta} \end{aligned} \right\} \Rightarrow \begin{aligned} q_i &= (O^T)_{\alpha i} (\tilde{q}_{\alpha} - a_{\alpha}) \\ q_j &= (O^T)_{\beta j} (\tilde{q}_{\beta} - a_{\beta}) \end{aligned}$$

Also we need to assume transformations are time-independent. (This will be clear in later steps). Then, velocities transform as:

$$\left. \begin{aligned} \dot{\tilde{q}}_{\alpha} &= O_{\alpha i} \dot{q}_i \\ \dot{\tilde{q}}_{\beta} &= O_{\beta j} \dot{q}_j \end{aligned} \right\} \Rightarrow \begin{aligned} \dot{q}_i &= (O^T)_{\alpha i} \dot{\tilde{q}}_{\alpha} \\ \dot{q}_j &= (O^T)_{\beta j} \dot{\tilde{q}}_{\beta} \end{aligned}$$

For orthonormal matrices:

$$O O^T = I \Rightarrow O^T = O^{-1}$$

Also $O_{ij} = (O^T)_{ji}$ for any matrix

Then our initial lagrangian becomes:

$$L = \frac{1}{2} \sum_i \sum_j (O^T)_{\alpha i} M_{ij} O_{\beta j} \dot{\tilde{q}}_{\alpha} \dot{\tilde{q}}_{\beta} - \frac{1}{2} \sum_i \sum_j (O^T)_{\alpha i} V_{ij} O_{\beta j} (\tilde{q}_{\alpha} - a_{\alpha}) (\tilde{q}_{\beta} - a_{\beta})$$

Let's focus on the first term. One can notice that, the first term of the new lagrangian is actually in the form:

$$m_{\alpha} \dot{\tilde{q}}_{\alpha} \dot{\tilde{q}}_{\alpha} = \delta_{\alpha\beta} m_{\alpha\beta} \dot{\tilde{q}}_{\alpha} \dot{\tilde{q}}_{\beta} = m_{\alpha\beta} \dot{\tilde{q}}_{\alpha} \dot{\tilde{q}}_{\beta}$$

Then surely, $M_{\alpha\beta}$ must be a diagonal matrix. That is

$$M_{\alpha\beta} = (O^T)_{\alpha i} M_{ij} O_{j\beta} \quad \text{actually represent a diagonalization process.}$$

→ We're actually familiar with this concept from moment of inertia tensor. We know that a real symmetric matrix (namely moment of inertia tensor) can be diagonalized via similarity transformation using a specific orthonormal matrix. That said orthonormal matrix has columns formed by the eigenvectors of the said matrix:

$$I'_{\alpha\beta} = O_{\alpha\alpha} I_{\alpha\beta} O_{\beta\alpha}^T \longrightarrow \begin{array}{l} \text{columns formed by eigenvectors} \\ \text{of } I_{\alpha\beta} \end{array}$$

diagonal entries are the eigenvalues of I

Then $\underbrace{m_{\alpha} = \delta_{\alpha\beta} M_{\alpha\beta}}_{\text{eigenvalues of } M}$ where $\underbrace{M_{\alpha\beta} = (O^T)_{\alpha i} M_{ij} O_{j\beta}}_{O \text{ is the orthogonal matrix whose columns are the eigenvectors of } M.}$

Now for the second term we have:

$$- \frac{1}{2} \sum_i \sum_j (O^T)_{\alpha i} V_{ij} O_{j\beta} (\tilde{q}_{\alpha} - a_{\alpha}) (\tilde{q}_{\beta} - a_{\beta})$$

If we look at the form of the new lagrangian, we conclude that $a_{\alpha} = a_{\beta} = 0$. New lagrangian then becomes

$$L = \frac{1}{2} \sum_{\alpha} m_{\alpha} (\dot{\tilde{q}}_{\alpha})^2 - \frac{1}{2} \sum_{\alpha} \sum_{\beta} \underbrace{(O^T)_{\alpha i} V_{ij} O_{j\beta}}_{\tilde{V}_{\alpha\beta}} \tilde{q}_{\alpha} \tilde{q}_{\beta}$$

Lastly we should investigate if $\tilde{V}$ is symmetric:

$$\tilde{V}_{\alpha\beta} = (O^T)_{\alpha i} V_{ij} O_{j\beta}$$

Change the indices $i \leftrightarrow j$ and $\alpha \leftrightarrow \beta$ since they are arbitrary

$$\tilde{V}_{\beta\alpha} = (O^T)_{\beta j} V_{ji} O_{i\alpha}$$

Since V is symmetric $V_{ji} = V_{ij}$ and also $O_{i\alpha} = (O^T)_{\alpha i}$:

$$\tilde{V}_{\beta\alpha} = (O^T)_{\alpha i} V_{ij} O_{j\beta} = \tilde{V}_{\alpha\beta} \quad \checkmark$$

All the relations are confirmed and thus, have a valid transformation of general coordinates:

O = eigen vector matrix of M OR $O_{\alpha\beta} = \left[\text{Null}(M - m_\beta I) \right]_\alpha$

$\tilde{q}_\alpha = O_{\alpha i} q_i$ $M_\alpha = \delta_{\alpha\beta} (O^T)_{\alpha i} M_{ij} O_{j\beta}$ OR eigen values of M

$$\tilde{V}_{\alpha\beta} = (O^T)_{\alpha i} V_{ij} O_{j\beta}$$

6)

Lagrangian for a charged particle of mass m & charge q is:

$$L = \frac{1}{2} m \dot{\vec{r}}^2 - qV + q \dot{\vec{r}} \cdot \frac{\vec{A}}{c}$$

$$\dot{\vec{r}}^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2$$

$$\vec{v} = \dot{x} \hat{x} + \dot{y} \hat{y} + \dot{z} \hat{z}$$

→ Here $V(x, y, z, t)$ is defined as electrostatic potential. For time independent magnetic field it can basically be expressed as:

$$V = - \int \vec{E} \cdot d\vec{s} \quad \vec{E} = -\vec{\nabla} V$$

→ $\vec{A}(x, y, z, t)$ is defined as magnetic vector potential. But how can we just define a potential, is it pertinent with Maxwell's equations?

$$\vec{\nabla} \cdot \vec{B} = 0 \quad \text{and we know that } \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0 \quad \text{for any } \vec{A}$$

Then we can define $\vec{A}$ s.t. $\vec{B} = \vec{\nabla} \times \vec{A}$

→ Maxwell's 2nd equation is in it's most general form:

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \text{but for } \vec{E} = -\vec{\nabla} V \Rightarrow \vec{\nabla} \times \vec{E} = -\vec{\nabla} \times (\vec{\nabla} V) = 0$$

Then we must have additional terms in the expression of $\vec{\nabla} V$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} = - \frac{\partial (\vec{\nabla} \times \vec{A})}{\partial t} \Rightarrow \vec{E} = -\vec{\nabla} V - \frac{\partial \vec{A}}{\partial t}$$

Now let's calculate $\frac{\partial L}{\partial \dot{q}}$, $\frac{\partial L}{\partial q}$. Since there are no constraints

on our system x, y, z forms a set of generalized coordinates.

We will write Euler-Lagrangian equations for each x, y, z coordinates separately

$$\frac{\partial L}{\partial \dot{\mathbf{x}}} = m \dot{\mathbf{x}} + q \frac{\partial \vec{V}}{\partial \dot{\mathbf{x}}} \cdot \vec{A} = m \dot{\mathbf{x}} + q \hat{\mathbf{x}} \cdot \vec{A}$$

* V and $\vec{A}$ are independent of velocities.

$$\frac{\partial L}{\partial \mathbf{x}} = -q \frac{\partial V}{\partial \mathbf{x}} + q \vec{V} \cdot \frac{\partial \vec{A}}{\partial \mathbf{x}}, \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{x}}} \right) = m \ddot{\mathbf{x}} + q \hat{\mathbf{x}} \cdot \frac{\partial \vec{A}}{\partial t}$$

Euler-Lagrange equations will read as:

$$m \ddot{x} = -q \frac{\partial V}{\partial x} - q \hat{x} \cdot \frac{\partial \vec{A}}{\partial t} + q \vec{V} \cdot \frac{\partial \vec{A}}{\partial x} \quad / \hat{x}$$

$$m \ddot{y} = -q \frac{\partial V}{\partial y} - q \hat{y} \cdot \frac{\partial \vec{A}}{\partial t} + q \vec{V} \cdot \frac{\partial \vec{A}}{\partial y} \quad / \hat{y}$$

$$m \ddot{z} = -q \frac{\partial V}{\partial z} - q \hat{z} \cdot \frac{\partial \vec{A}}{\partial t} + q \vec{V} \cdot \frac{\partial \vec{A}}{\partial z} \quad / \hat{z}$$

+

$$m (\ddot{x} \hat{x} + \ddot{y} \hat{y} + \ddot{z} \hat{z}) = -q \underbrace{\left(\frac{\partial V}{\partial x} \hat{x} + \frac{\partial V}{\partial y} \hat{y} + \frac{\partial V}{\partial z} \hat{z} \right)}_{\nabla V} - q \underbrace{\left(\frac{\partial A_x}{\partial t} \hat{x} + \frac{\partial A_y}{\partial t} \hat{y} + \frac{\partial A_z}{\partial t} \hat{z} \right)}_{\frac{\partial \vec{A}}{\partial t}} + q \underbrace{\left[\left(\vec{V} \cdot \frac{\partial \vec{A}}{\partial x} \right) \hat{x} + \left(\vec{V} \cdot \frac{\partial \vec{A}}{\partial y} \right) \hat{y} + \left(\vec{V} \cdot \frac{\partial \vec{A}}{\partial z} \right) \hat{z} \right]}_{\vec{C}}$$

Now let's try to figure out what the last term $\vec{C}$ is:

$$\vec{C} = \sum_{i=1}^3 \left(\vec{V} \cdot \frac{\partial \vec{A}}{\partial q_i} \right) \hat{e}_i, \quad C_i = \sum_{i=1}^3 \left(\vec{V} \cdot \frac{\partial \vec{A}}{\partial q_i} \right); \quad \text{where } q_1 = x, q_2 = y, q_3 = z$$

$$\text{One can write } \vec{V} \cdot \frac{\partial \vec{A}}{\partial q_i} = \frac{\partial}{\partial q_i} (\vec{V} \cdot \vec{A}) - \frac{\partial \vec{V}}{\partial q_i} \cdot \vec{A}$$

$$\begin{aligned} \vec{C} &= \sum_{i=1}^3 \frac{\partial}{\partial q_i} (\underbrace{\vec{V} \cdot \vec{A}}_{\text{scalar}}) \hat{e}_i - \underbrace{\left(\frac{\partial \vec{V}}{\partial q_i} \cdot \vec{A} \right)}_{\text{scalar}} \hat{e}_i \Rightarrow \\ &= \sum_{i=1}^3 \frac{\partial}{\partial q_i} (\vec{V} \cdot \vec{A}) \hat{e}_i - \vec{A} \sum_{i=1}^3 \left(\frac{\partial \vec{V}}{\partial q_i} \right) \hat{e}_i \end{aligned}$$

Last transformation is not so trivial but can be confirmed by following:

$$\begin{aligned} \sum_{i=1}^3 \left(\frac{\partial \vec{V}}{\partial q_i} \cdot \vec{A} \right) \hat{e}_i &= \sum_{i=1}^3 \sum_{j=1}^3 \frac{\partial V_j}{\partial q_i} A_j \hat{e}_i = \sum_{j=1}^3 A_j \underbrace{\sum_{i=1}^3 \frac{\partial V_j}{\partial q_i} \hat{e}_i}_{(\nabla \cdot \vec{V}) \delta_{ij}} \\ &= \sum_j A_j (\nabla \cdot \vec{V}) \delta_{ij} \hat{e}_i = \sum_j A_j \hat{e}_j (\nabla \cdot \vec{V}) = \vec{A} (\nabla \cdot \vec{V}) \end{aligned}$$

Then $\vec{C}$ transform into:

$$\vec{C} = \nabla(\vec{V} \cdot \vec{A}) - \vec{A}(\nabla \cdot \vec{V})$$

Recalling the property $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$ one can write:

$$\vec{C} = \vec{V} \times (\nabla \times \vec{A}) = \vec{V} \times \vec{B}$$

If we go back to the Euler-Lagrange equations:

$$m(\ddot{x}\hat{x} + \ddot{y}\hat{y} + \ddot{z}\hat{z}) = m\vec{a} = \vec{F} = q \underbrace{\left(-\vec{\nabla}V - \frac{\partial \vec{A}}{\partial t} \right)}_{\vec{E}} + q \vec{V} \times \vec{B}$$

$$\vec{F} = q(\vec{E} + \vec{V} \times \vec{B})$$

Lagrangian of a particle in an electromagnetic potential field yields the Lorentz force law as expected.